

Quiz (Carolina Reaper)

- Q1.** What is the primary purpose of a box-and-whisker plot?
(A) To display the mean and median
(B) To show the distribution of data
(C) To calculate the standard deviation
(D) To identify the mode
(E) To create a histogram
- Q2.** Which part of the box-and-whisker plot represents the interquartile range?
(A) The box
(B) The whiskers
(C) The median line
(D) The outliers
(E) The axes
- Q3.** What is the formula for the interquartile range (IQR)?
(A) $IQR = Q_3 - Q_1$
(B) $IQR = Q_1 - Q_3$
(C) $IQR = Q_2 - Q_1$
(D) $IQR = Q_3 - Q_2$
(E) $IQR = Q_1 + Q_3$
- Q4.** Which of the following is an example of an outlier in a dataset?
(A) A data point that is 1 unit away from the median
(B) A data point that is $1.5 \cdot IQR$ away from the median
(C) A data point that is 3 units away from the mean
(D) A data point that is equal to the mode
(E) A data point that is equal to the median
- Q5.** What is the first quartile (Q_1) in a dataset?
(A) The median of the dataset
(B) The mean of the dataset
(C) The lowest value in the dataset
(D) The median of the lower half of the dataset
(E) The highest value in the dataset
- Q6.** Which of the following statements about box-and-whisker plots is true?
(A) The plot is only used for small datasets
(B) The plot is only used for large datasets
(C) The plot can be used for both small and large datasets
(D) The plot is used to display categorical data
(E) The plot is used to display time-series data
- Q7.** How do you calculate the third quartile (Q_3) in a dataset?
(A) Find the median of the upper half of the dataset
(B) Find the median of the lower half of the dataset
(C) Find the mean of the dataset
(D) Find the mode of the dataset
(E) Find the range of the dataset
- Q8.** Which of the following is a benefit of using a box-and-whisker plot?
(A) It can only be used for one variable
(B) It can only be used for two variables
(C) It can be used to compare multiple variables
(D) It can only be used for continuous data
(E) It can only be used for discrete data

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the dataset: 2, 4, 6, 8, 10, 12, 14, 16, 100. Identify the quartiles, interquartile range, and any outliers.
- Q10.** A dataset has the following box-and-whisker plot: $Q_1 = 10$, $Q_3 = 30$, median = 20. If a new data point is added that is 50 units away from the median, determine if it is an outlier. Justify your answer.

Chef's Tips

- Tip 1: Make sure to check for any calculation errors
- Tip 2: Verify that the box-and-whisker plots are correctly constructed
- Tip 3: Ensure that the students provide clear and concise justifications for their answers